SAHITHI K

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PROFESSIONAL SUMMARY

Master's student in Health Data Science with hands-on experience in predictive modeling, machine learning, and data visualization. Proficient in Python, SQL, and R, with a solid foundation in healthcare analytics. Demonstrated ability to apply statistical models to real-world datasets, including heart disease and insurance rate analysis. Eager to contribute to healthcare innovation through data-driven insights.

SKILLS

- Programming: Python, R, SQL
- Data Science: Regression (Linear, Logistic), Classification, Clustering, NLP, SMOTE, SHAP, TF-IDF, Feature Engineering
- Tools: Tableau, Power BI, Scikit-learn, Pandas, NumPy, SciSpaCy, Matplotlib, Seaborn
- Platforms: Azure, BigQuery, Snowflake
- Databases: SQL Server, MySQL

EDUCATION

Master of Science in Health Data Science — **August2023 – August 2025**

Saint louis university, Saint louis, MO — ***CGPA – 3.8/4***

Bachelor of Pharmacy — **September2019 - May2023**

Sir C R Reddy college of pharmaceutical sciences, Eluru, AP — ***CGPA –8.00/10***

PROJECTS

Pregnancy Drug Safety Prediction (Capstone Project)

- Developed a **supervised machine learning and NLP framework** to predict FDA pregnancy drug safety categories using structured attributes and unstructured side-effect narratives.
- Applied **text preprocessing (SciSpaCy, TF-IDF, lemmatization)** and **one-hot encoding** for categorical data; performed Chi-square and ANOVA tests to identify significant predictors.
- Built and optimized multiple ML models (Logistic Regression, SVM, KNN, Decision Trees, Random Forest, Gradient Boosting, XGBoost, ANN); achieved **90.9% accuracy, F1 = 0.92, AUC = 0.98** with a SMOTE-enhanced ANN.
- Implemented **SHAP analysis** and **topic modeling (NMF)** for interpretability, uncovering clinically relevant side-effect clusters aligned with pregnancy risks.

Heart Disease Prediction using High-Performance Computing (HPC)

- Implemented logistic regression and random forest models on the *heart disease* dataset (Cleveland, Hungary, Switzerland, Long Beach V).
- Performed performance profiling in R, demonstrating fread was ~40x faster than read.csv while maintaining identical results.
- Applied **parallel processing** with joblib and optimized runtime using **Dask** for distributed computing.
- Achieved **98% model accuracy** with strong precision, recall, and F1-score; identified cholesterol and age as key predictors.
- Conducted advanced data analysis (cholesterol trends by age/sex) to derive health insights supporting targeted interventions.

Heart Disease Prediction using Machine Learning Models

- Conducted **exploratory data analysis (EDA)** including bar charts, histograms, box plots, and scatter plots to examine demographic and clinical predictors.
- Built and compared classification models (**K-Nearest Neighbors, Naïve Bayes, SVM, Random Forest**) for disease prediction.
- Optimized **KNN classifier** (k=5–10) achieving up to **99% accuracy**, outperforming linear models and demonstrating robustness against overfitting.
- Evaluated models using confusion matrices, accuracy, precision, recall, and F1-scores; identified trade-offs in predictive performance.
- Contributed to **data collection** and feature engineering, ensuring data quality for machine learning workflows.

Health Insurance Charges Analysis using Statistical Modeling

- Investigated correlations between **insurance costs, demographic factors (age, sex, region), BMI, and smoking behavior** using a U.S. health insurance dataset (Kaggle).
- Performed **data preprocessing with MySQL** and applied **statistical tests (t-tests, ANOVA, post-hoc analysis)** to identify significant differences across groups.
- Built a **linear regression model** achieving **R^2 = 74.75%**, confirming strong predictive power of age, BMI, and smoking status on healthcare costs.
- Created **data visualizations** (scatter plots, violin plots, box plots, 3D plots) to highlight disparities in charges across demographic and lifestyle factors.
- Found **smoking status as the strongest predictor** of higher healthcare charges, emphasizing the need for targeted interventions in public health policy.

Vision and Eye Health Surveillance System

- Enhanced analysis of **CDC's VEHSS database** by building Python functions to explore eye disease prevalence, demographics, and disparities.
- Extracted and cleaned large-scale public health datasets via **Socrata Open Data API (sodapy)**.
- Conducted **exploratory data analysis (EDA)** to evaluate prevalence of vision disorders and service utilization.
- Implemented methods for **categorical proportion analysis, prevalence estimation, and pivot-table reporting** across demographic and health variables.
- Integrated external datasets for **disease-specific analysis**, identifying groups disproportionately affected by vision conditions.